

Math Thermo
class # 23
4/21/2026

Application: Two-Phase Incompressible Flow

We will assume that

- 1) $r = 2$ (binary mixture),
- 2) $\psi_i \equiv 0$, $i = 1, 2$ (no external potential),
- 3) $\sigma_k \equiv 0$, $k = 1, \dots, K$ (no reactions),
- 4) $p \equiv p_0 \equiv \text{constant}$ (incompressible).

Since the mixture is incompressible, the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

is replaced by

$$\nabla \cdot \vec{v} = 0$$

The partial densities are

$$\rho_i = \rho_0 \phi_i, \quad i = 1, 2.$$

where

$$\phi_1 + \phi_2 = 1$$

are the mass fractions. We will assume that

$$U = \int_{\Omega} \hat{u}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) d\vec{x}$$

$$F = \int_{\Omega} \hat{f}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) d\vec{x}$$

$$S = \int_{\Omega} \hat{s}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) d\vec{x}$$

To shorten the writing, we will use the notation

$$\hat{u}(T, \phi, \nabla \phi) = \hat{u}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2).$$

We will assume (more on this later) that

$$(23.1) \quad \hat{f} = \hat{u} - T \hat{s}$$

as in equilibrium. This is another consequence of local thermodynamic equilibrium.

Further, we assume, for simplicity,

$$p_u =: \hat{u}(T, \phi, \nabla \phi) = \tilde{u}(T) + \lambda_u \delta(\phi, \nabla \phi),$$

$$p_f =: \hat{f}(T, \phi, \nabla \phi) = \tilde{f}(T) + \lambda_f \delta(\phi, \nabla \phi),$$

$$p_s =: \hat{s}(T, \phi, \nabla \phi) = \tilde{s}(T) + \lambda_s \delta(\phi, \nabla \phi).$$

λ_u and
 λ_s are
constants

In other words, the T and ϕ dependencies are separable. For $\tilde{u}, \tilde{f}, \tilde{s}$ we will eventually take

$$\tilde{u} = p_0 C_n T,$$

$$\tilde{f} = p_0 C_n T - p_0 C_n T \ln\left(\frac{T}{T_0}\right),$$

$$\tilde{s} = p_0 C_h \ln\left(\frac{T}{T_0}\right).$$

Clearly,

$$\tilde{f} = \tilde{u} - T\tilde{s}.$$

For the gradient term, we use the model

$$\delta(\phi, \nabla\phi) = \frac{1}{\varepsilon} \frac{\phi_1^2 \phi_2^2}{4} + \frac{\varepsilon}{2} \left(\frac{|\nabla\phi_1|^2}{2} + \frac{|\nabla\phi_2|^2}{2} \right),$$

where $\varepsilon > 0$ has units of length. $[\varepsilon] = \text{length}^{-1}$.

Finally, we note that, it must be true that

$$\lambda_f = \lambda_u - T\lambda_s.$$

Now, let us note that

$$\frac{\partial \tilde{s}}{\partial T} = p_0 C_h \frac{1}{T}$$

$$\frac{\partial \tilde{u}}{\partial T} = p_0 C_h$$

so that

(23.2)

$$\boxed{\frac{\partial \tilde{s}}{\partial T} = \frac{1}{T} \frac{\partial \tilde{u}}{\partial T} .}$$

This condition is called thermal compatibility. We will generally assume this holds.

Now, we are going to derive the entropy production terms in a slightly different way. Rather than relying on a Gibbs relation, we will directly calculate the RHS of

$$\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) = -\nabla \cdot \vec{J}_S + \sigma_S$$

in order to determine $\vec{J}_S$ and σ_S .

Later on we will explain the rationale for this procedure. Before we do this, here are the governing equations:

$$(23.3) \quad \nabla \cdot \vec{v} = 0,$$

$$(23.4) \quad \rho_0 \frac{\partial \phi_i}{\partial t} + \rho_0 \nabla \cdot (\phi_i \vec{v}) = -\nabla \cdot \vec{J}_i^D, \quad (i=1,2)$$

$$(23.5) \quad \rho_0 \frac{\partial \vec{v}}{\partial t} + \rho_0 \nabla \cdot (\vec{v} \otimes \vec{v}) + \nabla P = -\nabla \cdot \Pi,$$

$$(23.6) \quad \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) = -\nabla \cdot \vec{J}_U^D - \Pi : \nabla \vec{v},$$

where we have used $\psi_i \equiv 0$, $\vec{\sigma}_K \equiv 0$, $\nabla \cdot \vec{v} \equiv 0$, $\rho_0 \equiv \text{const}$, et cetera.

We must find Π , $\vec{J}_i^D$, $\vec{J}_U^D$ so that entropy is always increasing. We need some preliminary calculations.

Proposition (23.1): let us assume that

$$\rho u =: \hat{u} = \tilde{u}(\tau) + \lambda_u \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2).$$

Then, if $\vec{v}$ is the barycentric velocity, we have

$$\frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) = \frac{\partial \tilde{u}}{\partial \tau} \left(\frac{\partial \tau}{\partial t} + \nabla \cdot (\tau \vec{v}) \right)$$

$$\mathcal{S} := \frac{1}{4\varepsilon} \phi_1^2 \phi_2^2 + \frac{\varepsilon}{2} \left(\frac{|\nabla \phi_1|^2}{2} + \frac{|\nabla \phi_2|^2}{2} \right)$$

(23.7)

$$\begin{aligned} & + \lambda_u \omega_2 \left(\frac{\partial \phi_2}{\partial t} + \nabla \cdot (\phi_2 \vec{v}) \right) \\ & + \lambda_u \omega_1 \left(\frac{\partial \phi_1}{\partial t} + \nabla \cdot (\phi_1 \vec{v}) \right) \\ & + \frac{\lambda_u}{2} \varepsilon \nabla \cdot \left(\nabla \phi_2 \frac{\partial \phi_2}{\partial t} \right) \\ & + \frac{\lambda_u}{2} \varepsilon \nabla \cdot \left(\nabla \phi_1 \frac{\partial \phi_1}{\partial t} \right) \\ & + \frac{\lambda_u}{2} \varepsilon \left[\nabla \cdot (\nabla \phi_2 \otimes \nabla \phi_2) \right] \cdot \vec{v} \\ & + \frac{\lambda_u}{2} \varepsilon \left[\nabla \cdot (\nabla \phi_1 \otimes \nabla \phi_1) \right] \cdot \vec{v}, \end{aligned}$$

where

$$\omega_i = \frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_i} - \frac{\varepsilon}{2} \Delta \phi_i, \quad i=1,2,$$

and

$$W(\phi_1, \phi_2) = \frac{1}{4} \phi_1^2 \phi_2^2.$$

Proof: We use the general form

$$\rho u = \hat{u} = \tilde{u}(\tau) + \lambda_u \left(\frac{1}{\varepsilon} W(\phi_1, \phi_2) + \frac{\varepsilon}{2} \left(\frac{|\nabla \phi_1|^2}{2} + \frac{|\nabla \phi_2|^2}{2} \right) \right).$$

Now,

$$\begin{aligned} \frac{\partial u}{\partial t} &= \frac{\partial \tilde{u}}{\partial \tau} \frac{\partial \tau}{\partial t} + \lambda_u \left(\frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_1} \frac{\partial \phi_1}{\partial t} + \frac{\partial W}{\partial \phi_2} \frac{\partial \phi_2}{\partial t} \right. \\ &\quad \left. + \frac{\varepsilon}{2} \nabla \phi_1 \cdot \nabla \left(\frac{\partial \phi_1}{\partial t} \right) + \frac{\varepsilon}{2} \nabla \phi_2 \cdot \nabla \left(\frac{\partial \phi_2}{\partial t} \right) \right) \end{aligned}$$

Next,

$$\begin{aligned} \nabla \cdot (\hat{u} \vec{v}) &= \nabla \cdot (\tilde{u}(\tau) \vec{v}) + \frac{\lambda_u}{\varepsilon} \nabla \cdot (W(\phi_1, \phi_2) \vec{v}) \\ &\quad + \frac{\lambda_u \varepsilon}{4} \nabla \cdot (|\nabla \phi_1|^2 \vec{v}) + \frac{\lambda_u \varepsilon}{4} \nabla \cdot (|\nabla \phi_2|^2 \vec{v}). \end{aligned}$$

Notice that,

$$\begin{aligned}
 \nabla \cdot (\tilde{u}(\tau) \vec{v}) &= \nabla(\tilde{u}(\tau)) \cdot \vec{v} + \tilde{u}(\tau) \nabla \cdot \vec{v} \\
 &= \frac{\partial \tilde{u}}{\partial \tau} \nabla \tau \cdot \vec{v} \\
 &= \frac{\partial \tilde{u}}{\partial \tau} \nabla \cdot (\tau \vec{v}).
 \end{aligned}$$

likewise,

$$\begin{aligned}
 \nabla \cdot (W(\phi_1, \phi_2) \vec{v}) &= \nabla(W(\phi_1, \phi_2)) \cdot \vec{v} + W(\phi_1, \phi_2) \nabla \cdot \vec{v} \\
 &= \frac{\partial W}{\partial \phi_1} \nabla \phi_1 \cdot \vec{v} + \frac{\partial W}{\partial \phi_2} \nabla \phi_2 \cdot \vec{v} \\
 &= \frac{\partial W}{\partial \phi_1} \nabla \cdot (\phi_1 \vec{v}) + \frac{\partial W}{\partial \phi_2} \nabla \cdot (\phi_2 \vec{v})
 \end{aligned}$$

Finally, consider the following calculation: using Einstein's summation convention,

$$\begin{aligned}
 \nabla \cdot (|\nabla \phi|^2 \vec{v}) &= \frac{\partial}{\partial x_i} \left(\frac{\partial \phi}{\partial x_j} \frac{\partial \phi}{\partial x_j} v_i \right) \\
 &= \frac{\partial^2 \phi}{\partial x_i \partial x_j} \frac{\partial \phi}{\partial x_j} v_i + \frac{\partial \phi}{\partial x_j} \frac{\partial^2 \phi}{\partial x_i \partial x_j} v_i \\
 &\quad + \frac{\partial \phi}{\partial x_j} \frac{\partial \phi}{\partial x_j} \frac{\partial v_i}{\partial x_i} \\
 &= 2 \frac{\partial \phi}{\partial x_j} \frac{\partial^2 \phi}{\partial x_i \partial x_j} v_i.
 \end{aligned}$$

But note that

$$\begin{aligned}
 \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v} &= \frac{\partial}{\partial x_j} \left(\frac{\partial \phi}{\partial x_i} \frac{\partial \phi}{\partial x_j} \right) v_i \\
 &= \frac{\partial^2 \phi}{\partial x_i \partial x_j} \frac{\partial \phi}{\partial x_j} v_i + \frac{\partial \phi}{\partial x_i} v_i \frac{\partial^2 \phi}{\partial x_j \partial x_j} \\
 &= \frac{1}{2} \nabla \cdot (|\nabla \phi|^2 \vec{v}) \\
 &\quad + \vec{v} \cdot \nabla \phi \Delta \phi.
 \end{aligned}$$

Thus,

$$\nabla \cdot (|\nabla \phi|^2 \vec{v}) = 2 \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v} - 2 \vec{v} \cdot \nabla \phi \Delta \phi$$

Putting everything together,

$$\begin{aligned} \frac{\partial u}{\partial t} + \nabla \cdot (u \vec{v}) &= \frac{\partial \tilde{u}}{\partial T} \frac{\partial T}{\partial t} + \lambda_u \left(\frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_1} \frac{\partial \phi_1}{\partial t} + \frac{\partial W}{\partial \phi_2} \frac{\partial \phi_2}{\partial t} \right. \\ &\quad \left. + \frac{\varepsilon}{2} \nabla \phi_1 \cdot \nabla \left(\frac{\partial \phi_1}{\partial t} \right) + \frac{\varepsilon}{2} \nabla \phi_2 \cdot \nabla \left(\frac{\partial \phi_2}{\partial t} \right) \right) \\ &\quad + \frac{\partial \tilde{u}}{\partial T} \nabla \cdot (T \vec{v}) \\ &\quad + \frac{\lambda_u}{\varepsilon} \left(\frac{\partial W}{\partial \phi_1} \nabla \cdot (\phi_1 \vec{v}) + \frac{\partial W}{\partial \phi_2} \nabla \cdot (\phi_2 \vec{v}) \right) \\ &\quad + \frac{\lambda_u \varepsilon}{2} \left[\nabla \cdot (\nabla \phi_1 \otimes \nabla \phi_1) \right] \cdot \vec{v} \\ &\quad + \frac{\lambda_u \varepsilon}{2} \left[\nabla \cdot (\nabla \phi_2 \otimes \nabla \phi_2) \right] \cdot \vec{v} \\ &\quad - \frac{\lambda_u \varepsilon}{2} \Delta \phi_1 \nabla \cdot (\phi_1 \vec{v}) \\ &\quad - \frac{\lambda_u \varepsilon}{2} \Delta \phi_2 \nabla \cdot (\phi_2 \vec{v}). \end{aligned}$$

We use the identity

$$\nabla \phi \cdot \nabla \left(\frac{\partial \phi}{\partial t} \right) = \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) - \Delta \phi \frac{\partial \phi}{\partial t}$$

and some simplification to obtain

$$\begin{aligned}
\frac{\partial u}{\partial t} + \nabla \cdot (u \vec{v}) &= \frac{\partial \tilde{u}}{\partial T} \frac{\partial T}{\partial t} + \lambda_u \left(\frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_1} \frac{\partial \phi_1}{\partial t} + \frac{\partial W}{\partial \phi_2} \frac{\partial \phi_2}{\partial t} \right. \\
&\quad + \frac{\varepsilon}{2} \nabla \cdot \left(\nabla \phi_1 \frac{\partial \phi_1}{\partial t} \right) - \frac{\varepsilon}{2} \Delta \phi_1 \frac{\partial \phi_1}{\partial t} \\
&\quad + \frac{\varepsilon}{2} \nabla \cdot \left(\nabla \phi_2 \frac{\partial \phi_2}{\partial t} \right) - \frac{\varepsilon}{2} \Delta \phi_2 \frac{\partial \phi_2}{\partial t} \Big) \\
&\quad + \frac{\partial \tilde{u}}{\partial T} \nabla \cdot (T \vec{v}) \\
&\quad + \frac{\lambda_u}{\varepsilon} \left(\frac{\partial W}{\partial \phi_1} \nabla \cdot (\phi_1 \vec{v}) + \frac{\partial W}{\partial \phi_2} \nabla \cdot (\phi_2 \vec{v}) \right) \\
&\quad + \frac{\lambda_u \varepsilon}{2} \left[\nabla \cdot (\nabla \phi_1 \otimes \nabla \phi_1) \right] \cdot \vec{v} \\
&\quad + \frac{\lambda_u \varepsilon}{2} \left[\nabla \cdot (\nabla \phi_2 \otimes \nabla \phi_2) \right] \cdot \vec{v} \\
&\quad - \frac{\lambda_u \varepsilon}{2} \Delta \phi_1 \nabla \cdot (\phi_1 \vec{v}) \\
&\quad - \frac{\lambda_u \varepsilon}{2} \Delta \phi_2 \nabla \cdot (\phi_2 \vec{v}) \\
&= \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\
&\quad + \lambda_u \omega_1 \left(\frac{\partial \phi_1}{\partial t} + \nabla \cdot (\phi_1 \vec{v}) \right) \\
&\quad + \lambda_u \omega_2 \left(\frac{\partial \phi_2}{\partial t} + \nabla \cdot (\phi_2 \vec{v}) \right) \\
&\quad + \frac{\lambda_u \varepsilon}{2} \nabla \cdot \left(\nabla \phi_1 \frac{\partial \phi_1}{\partial t} \right) \\
&\quad + \frac{\lambda_u \varepsilon}{2} \nabla \cdot \left(\nabla \phi_2 \frac{\partial \phi_2}{\partial t} \right) \\
&\quad + \frac{\lambda_u \varepsilon}{2} \left[\nabla \cdot (\nabla \phi_1 \otimes \nabla \phi_1) \right] \cdot \vec{v} \\
&\quad + \frac{\lambda_u \varepsilon}{2} \left[\nabla \cdot (\nabla \phi_2 \otimes \nabla \phi_2) \right] \cdot \vec{v} //
\end{aligned}$$

Proposition (23.2): Recall that, since $r=2$,

$$\phi_1 + \phi_2 = 1.$$

As a consequence, setting

$$\phi := \phi_2,$$

it follows that

(23.8)

$$\begin{aligned} \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) &= \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\ &+ \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ \lambda_u \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ \lambda_u \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}, \end{aligned}$$

where

$$\omega := \frac{1}{\varepsilon} \tilde{W}'(\phi) - \varepsilon \Delta \phi, \quad \tilde{W}(\phi) := W(1-\phi, \phi).$$

Proof: Clearly,

$$\frac{\partial \phi_1}{\partial t} + \nabla \cdot (\phi_1 \vec{v}) = - \frac{\partial \phi}{\partial t} - \nabla \cdot (\phi \vec{v})$$

$$\frac{\partial \phi_2}{\partial t} + \nabla \cdot (\phi_2 \vec{v}) = \frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v})$$

$$\begin{aligned} I &:= \lambda_u \omega_1 \left(\frac{\partial \phi_1}{\partial t} + \nabla \cdot (\phi_1 \vec{v}) \right) + \lambda_u \omega_2 \left(\frac{\partial \phi_2}{\partial t} + \nabla \cdot (\phi_2 \vec{v}) \right) \\ &= \lambda_u (\omega_1 - \omega_2) \left(\frac{\partial \phi}{\partial t} - \nabla \cdot (\phi \vec{v}) \right) \\ &= \lambda_u \left(\frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_1} - \frac{1}{\varepsilon} \frac{\partial W}{\partial \phi_2} - \frac{\varepsilon}{2} \Delta \phi_1 + \frac{\varepsilon}{2} \Delta \phi_2 \right) \times \end{aligned}$$

$$\left(\frac{\partial \phi}{\partial t} - \nabla \cdot (\phi \vec{v}) \right)$$

where

$$= \lambda_u \left(\frac{1}{\varepsilon} \frac{\partial \tilde{W}}{\partial \phi} - \varepsilon \Delta \phi \right) \times \left(\frac{\partial \phi}{\partial t} - \nabla \cdot (\phi \vec{v}) \right)$$

$$\tilde{W}(\phi) := W(1-\phi, \phi).$$

The rest of the details are left for an exercise. ///

The entropy equation will have the form

$$\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) = - \nabla \cdot \mathbf{J}_S^D + \sigma_S.$$

We calculated the terms of this equation previously using the Gibbs relation

$$ds = \frac{1}{T} du + \frac{P}{T} dv - \sum_{i=1}^r \frac{\mu_i}{T} d\phi_i,$$

where s is the specific entropy, u is the specific internal energy density, v is the specific volume.

This equation might not hold any longer, since

$$S_v(t) = \int_{v(t)} \hat{S}(\tau, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) d\vec{x}$$

rather than

$$S_v(t) = \int_{v(t)} \rho s(u, v, \phi_1, \phi_2) d\vec{x}.$$

In the present case, we have a more straight -

forward way of finding J_s^D and σ_s .

Proposition (23.3): Suppose that thermal compatibility holds, ie,

$$(23.9) \quad \frac{\partial \tilde{S}}{\partial T} = \frac{1}{T} \frac{\partial \tilde{u}}{\partial T}$$

and assume that

$$(23.10) \quad p_u =: \hat{u}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{u}(T) + \lambda_u \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

$$(23.11) \quad p_f =: \hat{f}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{f}(T) + \lambda_f \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

$$(23.12) \quad p_s =: \hat{s}(T, \phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2) = \tilde{s}(T) + \lambda_s \delta(\phi_1, \nabla \phi_1, \phi_2, \nabla \phi_2),$$

with

$$\tilde{f} = \tilde{u} - T \tilde{s} \quad \text{and} \quad \lambda_f = \lambda_u - T \lambda_s.$$

Then,

$$(23.13) \quad \frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) = - \nabla \cdot \vec{J}_s^D + \sigma_s,$$

where

$$(23.14) \quad \vec{J}_s^D = \frac{1}{T} \vec{J}_u^D - \frac{\lambda_f}{\rho_0} \frac{\omega}{T} \vec{J}_2^D + \lambda_f \varepsilon \frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} + \lambda_f \varepsilon \frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v}$$

$$(23.15) \quad \sigma_s = - \frac{1}{T} \nabla \vec{v} : \left(\Pi - \lambda_f \varepsilon \nabla \phi \otimes \nabla \phi \right) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D + \nabla \left(\frac{1}{T} \right) \cdot \left\{ \vec{J}_u^D + \lambda_f \varepsilon \nabla \phi \frac{\partial \phi}{\partial t} + \lambda_f \varepsilon (\nabla \phi \otimes \nabla \phi) \vec{v} \right\}$$

Proof: Using what we learned from Proposition (23.2), we must have

(23.16)

$$\begin{aligned} \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= \frac{\partial \tilde{S}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\ &+ \lambda_s \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ \lambda_s \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ \lambda_s \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}, \end{aligned}$$

where

$$\omega := \frac{1}{\varepsilon} \tilde{W}'(\phi) - \varepsilon \Delta \phi, \quad \tilde{W}(\phi) := W(1-\phi, \phi).$$

Recall, we have identified

$$\phi := \phi_2 \quad (\phi_1 = 1 - \phi).$$

Using thermal compatibility

$$\begin{aligned} \frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= \frac{1}{T} \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\ &+ \lambda_s \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\ &+ \lambda_s \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\ &+ \lambda_s \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}, \end{aligned}$$

or, equivalently,

$$\begin{aligned}
 (23.17) \quad T \left(\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) \right) &= \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\
 &+ T \lambda_s \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &+ T \lambda_s \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &+ T \lambda_s \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}$$

Using

$$(23.6) \quad \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) = -\nabla \cdot \vec{J}_U^0 - \Pi : \nabla \vec{v}$$

and

$$\begin{aligned}
 (23.8) \quad \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) &= \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) \\
 &+ \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &+ \lambda_u \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &+ \lambda_u \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}
 \end{aligned}$$

we have our heat equation:

$$\begin{aligned}
 (23.18) \quad \frac{\partial \tilde{u}}{\partial T} \left(\frac{\partial T}{\partial t} + \nabla \cdot (T \vec{v}) \right) &= -\nabla \cdot \vec{J}_U^0 - \Pi : \nabla \vec{v} \\
 &- \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &- \lambda_u \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &- \lambda_u \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}$$

Substituting (23.18) into (23.17) we have

$$\begin{aligned}
 T \left(\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) \right) &= -\nabla \cdot \vec{J}_0^D - \Pi : \nabla \vec{v} \\
 &\quad + (T \lambda_s - \lambda_u) \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &\quad + (T \lambda_s - \lambda_u) \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad + (T \lambda_s - \lambda_u) \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v} \\
 &= -\nabla \cdot \vec{J}_0^D - \Pi : \nabla \vec{v} \\
 &\quad - \lambda_f \omega \left(\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \vec{v}) \right) \\
 &\quad - \lambda_f \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad - \lambda_f \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}$$

Using

$$(23.4) \quad \rho_0 \frac{\partial \phi_i}{\partial t} + \rho_0 \nabla \cdot (\phi_i \vec{v}) = -\nabla \cdot \vec{J}_i^D$$

we have

$$\begin{aligned}
 T \left(\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) \right) &= -\nabla \cdot \vec{J}_0^D - \Pi : \nabla \vec{v} \\
 &\quad + \lambda_f \omega \frac{\nabla \cdot \vec{J}_2^D}{\rho_0} \\
 &\quad - \lambda_f \varepsilon \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &\quad - \lambda_f \varepsilon \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}$$

Equivalently,

$$\begin{aligned}
 \left(\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) \right) &= \frac{-\nabla \cdot \vec{J}_U^D}{T} - \frac{\Pi : \nabla \vec{v}}{T} \\
 &+ \frac{\lambda_f \omega}{\rho_0 T} \nabla \cdot \vec{J}_L^D \\
 &- \frac{\lambda_f \varepsilon}{T} \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) \\
 &- \frac{\lambda_f \varepsilon}{T} \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v}.
 \end{aligned}
 \tag{23.19}$$

Now, we use some identities:

$$\begin{aligned}
 \frac{\omega}{T} \nabla \cdot \vec{J}_L^D &= \nabla \cdot \left(\frac{\omega}{T} \vec{J}_L^D \right) - \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_L^D, \\
 \frac{1}{T} \nabla \cdot \vec{J}_U^D &= \nabla \cdot \left(\frac{1}{T} \vec{J}_U^D \right) - \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_U^D, \\
 \frac{1}{T} \nabla \cdot \left(\nabla \phi \frac{\partial \phi}{\partial t} \right) &= \nabla \cdot \left(\frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \right) - \nabla \left(\frac{1}{T} \right) \cdot \nabla \phi \frac{\partial \phi}{\partial t}, \\
 \left[\nabla \cdot (\nabla \phi \otimes \nabla \phi) \right] \cdot \vec{v} &= \nabla \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}] \\
 &\quad - (\nabla \phi \otimes \nabla \phi) : \nabla \vec{v}
 \end{aligned}$$

Inserting into (23.19), we find

$$\begin{aligned}
 \left(\frac{\partial \hat{S}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) \right) &= -\nabla \cdot \left(\frac{1}{T} \vec{J}_U^D \right) + \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_U^D \\
 &+ \frac{\lambda_f}{\rho_0} \nabla \cdot \left(\frac{\omega}{T} \vec{J}_L^D \right) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_L^D \\
 &- \lambda_f \varepsilon \nabla \cdot \left(\frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \right) + \lambda_f \varepsilon \nabla \left(\frac{1}{T} \right) \cdot \nabla \phi \frac{\partial \phi}{\partial t} \\
 &- \frac{\lambda_f \varepsilon}{T} \nabla \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}]
 \end{aligned}
 \tag{23.20}$$

$$+ \frac{\lambda_f \varepsilon}{T} (\nabla \phi \otimes \nabla \phi) : \nabla \vec{v} \quad - \frac{\Pi : \nabla \vec{v}}{T}$$

One more identity should do it :

$$\frac{1}{T} \nabla \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}] = \nabla \cdot \left[\frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v} \right] - \nabla \left(\frac{1}{T} \right) \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}].$$

Then (23.20) becomes

(23.21)

$$\begin{aligned} \frac{\partial \hat{\Sigma}}{\partial t} + \nabla \cdot (\hat{S} \vec{v}) &= -\nabla \cdot \left(\frac{1}{T} \vec{J}_v^D \right) + \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_v^D \\ &+ \frac{\lambda_f}{\rho_0} \nabla \cdot \left(\frac{\omega}{T} \vec{J}_2^D \right) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D \\ &- \lambda_f \varepsilon \nabla \cdot \left(\frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \right) + \lambda_f \varepsilon \nabla \left(\frac{1}{T} \right) \cdot \nabla \phi \frac{\partial \phi}{\partial t} \\ &+ \frac{\lambda_f \varepsilon}{T} (\nabla \phi \otimes \nabla \phi) : \nabla \vec{v} \quad - \frac{\Pi : \nabla \vec{v}}{T} \\ &- \lambda_f \varepsilon \nabla \cdot \left[\frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v} \right] \\ &+ \lambda_f \varepsilon \nabla \left(\frac{1}{T} \right) \cdot [(\nabla \phi \otimes \nabla \phi) \vec{v}]. \end{aligned}$$

Comparing (23.21) to (23.13), we find

$$\begin{aligned} J_s^D &= \frac{1}{T} J_v^D - \frac{\lambda_f}{\rho_0} \frac{\omega}{T} \vec{J}_2^D + \lambda_f \varepsilon \frac{1}{T} \nabla \phi \frac{\partial \phi}{\partial t} \\ &+ \lambda_f \varepsilon \frac{1}{T} (\nabla \phi \otimes \nabla \phi) \vec{v} \end{aligned}$$

$$\begin{aligned} \sigma_s &= -\frac{1}{T} \nabla \vec{v} : (\Pi - \lambda_f \varepsilon \nabla \phi \otimes \nabla \phi) - \frac{\lambda_f}{\rho_0} \nabla \left(\frac{\omega}{T} \right) \cdot \vec{J}_2^D \\ &+ \nabla \left(\frac{1}{T} \right) \cdot \left\{ J_v^D + \lambda_f \varepsilon \nabla \phi \frac{\partial \phi}{\partial t} + \lambda_f \varepsilon (\nabla \phi \otimes \nabla \phi) \vec{v} \right\} \end{aligned}$$

Thus (23.14) and (23.15) are confirmed. ///

Other forms are possible. Consider the following identity:

$$\nabla\left(\frac{\omega}{T}\right) \cdot \vec{J}_2^D = \frac{1}{T} \nabla \omega \cdot \vec{J}_2^D + \nabla\left(\frac{1}{T}\right) \cdot \omega \vec{J}_2^D.$$

Then,

$$\begin{aligned} (23.22) \quad \sigma_s = & -\frac{1}{T} \nabla \vec{v} : (\Pi - \lambda_f \varepsilon \nabla \phi \otimes \nabla \phi) - \frac{\lambda_f}{\rho_0 T} \nabla \omega \cdot \vec{J}_2^D \\ & + \nabla\left(\frac{1}{T}\right) \cdot \left\{ J_v^D + \lambda_f \varepsilon \nabla \phi \frac{\partial \phi}{\partial t} + \lambda_f \varepsilon (\nabla \phi \otimes \nabla \phi) \vec{v} \right. \\ & \left. - \frac{\lambda_f}{\rho_0} \omega \vec{J}_2^D \right\}. \end{aligned}$$